

# Notes: Schott (QJE, 2004)

## Across-Product versus Within-Product Specialization in International Trade

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# 1 Big picture

- **Question.** Do countries with different factor endowments specialize in different products, or do they specialize in different varieties *within* the same products?
- **Setting.** The paper studies U.S. product-level import data from 1972 to 1994. Each product-country pair is interpreted as a variety: for example, cotton shirts imported from Japan and cotton shirts imported from the Philippines are different varieties within the same product code.
- **Core empirical fact 1: little across-product specialization.** The United States increasingly imports the same narrowly defined products from both low-wage and high-wage countries. The share of products imported from both low- and high-wage countries rises strongly over time.
- **Core empirical fact 2: strong within-product specialization.** Within the same product category, unit values are systematically higher for varieties from high-income, capital-abundant, and skill-abundant countries.
- **Core empirical fact 3: prices track production techniques.** Within industries, countries that use more capital-intensive production techniques tend to export higher-unit-value varieties.
- **Old trade theory interpretation.** The evidence rejects a literal Heckscher-Ohlin prediction that countries specialize in completely different products, but it supports a more refined factor-proportions interpretation in which countries specialize in vertically differentiated varieties inside product categories.
- **New trade theory tension.** Standard new trade models with productivity-driven pricing predict that more productive producers should charge lower prices. If high-wage, capital-abundant countries are more productive, the observed high unit values of their exports are difficult to reconcile with those models.
- **Economic takeaway.** International specialization is partly hidden inside product categories. High-wage countries may remain in “low-wage” industries by producing higher-quality, more capital- or skill-intensive varieties, while low-wage countries produce lower-unit-value varieties.

# 2 Incremental contribution of the paper

- **From industry-level trade to product-level trade.** Earlier tests of factor-proportions trade often use industry-level data. This paper moves to much finer U.S. import product codes and shows that industry-level tests miss an important layer of specialization.
- **A new distinction: across-product versus within-product specialization.** The paper separates two ideas that are often conflated: whether countries export different product categories, and whether they export different varieties within the same product category.
- **Direct evidence on vertical differentiation.** By using unit values within narrow product codes, the paper provides evidence that high- and low-wage countries produce vertically differentiated goods rather than identical goods.
- **A bridge between old and new trade theory.** The paper uses the same product-level facts to evaluate both factor-proportions logic and new trade theory. It finds support for the former at the within-product level and tension with the latter’s inverse price-productivity prediction.

- **Dynamic contribution.** The paper shows that overlap between high- and low-wage exporters increases over time, while relative unit values also move with relative factor accumulation.
- **Measurement contribution.** The paper treats unit value dispersion not merely as noise, but as a potentially informative signal about product quality, attributes, and input intensity.
- **Implication for globalization.** The results help explain why high-wage countries continue to export in industries such as apparel and textiles: they may be exporting different, higher-unit-value varieties within those industries.

### 3 Data and measurement

#### 3.1 Product-level U.S. import data

- The paper uses U.S. import data compiled by Feenstra, covering customs values and quantities by product and exporting country from 1972 to 1994.
- The customs value is free-on-board value, excluding duties and shipping charges.
- A product-country unit value is defined as

$$u_{pc} = \frac{V_{pc}}{Q_{pc}},$$

where  $V_{pc}$  is import value and  $Q_{pc}$  is import quantity.

- Products are defined at very fine levels: seven-digit Tariff Schedule codes before 1989 and ten-digit Harmonized System codes from 1989 onward.
- The empirical analysis focuses on manufacturing imports, mainly SITC one-digit categories 5 through 8.

**Interpretation:** The key empirical object is not the industry, but the product-country variety. The paper asks whether Japan and the Philippines export the same product code, and if so, whether the Japanese variety has a higher unit value.

#### 3.2 Product, country, and variety definitions

- A product imported from different countries is treated as different varieties of the same product.
- A variety may be horizontally different, such as color or design, or vertically different, such as quality, technical sophistication, or added features.
- Products are aggregated into industries only for some descriptive and robustness exercises.
- The paper emphasizes that unit values cannot be computed cleanly at the industry level because quantity units differ across products.

**Interpretation:** The same product code can contain economically meaningful differences across source countries. Unit values are used to reveal these hidden differences.

### 3.3 Exporter income and factor endowments

- Exporter per capita GDP is used to classify countries into low-, middle-, and high-wage cohorts.
- In the baseline classification, low-wage countries are below the 30th percentile of world per capita GDP, middle-wage countries are between the 30th and 70th percentiles, and high-wage countries are above the 70th percentile.
- The paper also uses more direct factor-abundance measures: capital per labor and skill per labor.
- Capital per labor comes from the Penn World Tables. Skill per labor is based on educational attainment data from Barro and Lee.

**Interpretation:** Per capita GDP is a broad wage and development proxy. Capital per worker and skill per worker are closer to the Heckscher-Ohlin factors that motivate the theory.

### 3.4 Product cohorts for the across-product test

- Products are classified according to the exporter cohorts from which they originate.
- $L$ ,  $M$ , and  $H$  products come only from low-, middle-, or high-wage countries, respectively.
- $LM$  and  $MH$  products are imported from both adjacent groups.
- $LMH$  products are imported from at least one low-wage and at least one high-wage exporter in the same year.

**Interpretation:** A stark across-product factor-proportions prediction says low- and high-wage countries should not export many of the same products. A high or rising  $LMH$  share therefore works against across-product specialization.

### 3.5 Measurement limitations

- Unit values can reflect true quality differences, but they can also reflect measurement error, product misclassification, reporting errors, or differences in quantities and units.
- The paper acknowledges customs classification problems, but argues that the systematic relationships with income, factor abundance, production techniques, and time are difficult to explain as pure noise.
- Transfer pricing, bargaining power, demand-side taste differences, and outsourcing are discussed as alternative interpretations.

**Interpretation:** The results should be read as strong evidence of systematic within-product price variation, not as a perfect decomposition of price into quality and markup components.

## 4 Models and empirical specifications

### 4.1 Old trade theory benchmark

The paper begins from a Heckscher-Ohlin factor-proportions view. Countries with different relative factor endowments should specialize in goods whose production techniques match their factor abundance.

- Labor-abundant countries should export labor-intensive goods.
- Capital- and skill-abundant countries should export capital- and skill-intensive goods.
- In a strict across-product interpretation, high- and low-wage countries should export different product categories.
- In a within-product interpretation, the same product category can contain vertically distinct varieties produced with different input intensities.

**Interpretation:** The key innovation is to allow the isoquants in a Heckscher-Ohlin diagram to represent varieties inside product codes, not just broad industries.

## 4.2 New trade theory benchmark

In a simple new trade model with imperfect competition and productivity differences, price is a markup over productivity-adjusted marginal cost. For two varieties  $m$  and  $n$ ,

$$\frac{p_m}{p_n} = \frac{w/\phi_m}{w/\phi_n},$$

where  $w$  is the wage and  $\phi$  is productivity.

- Holding efficiency-adjusted wages fixed, the higher-productivity producer should charge a lower price.
- If high-wage, capital-abundant countries are also high-productivity countries, their varieties should not systematically have higher prices.
- The data instead show that high-income, capital-abundant, and skill-abundant exporters sell higher-unit-value varieties.

**Interpretation:** The paper does not claim that all new trade theory is wrong. Rather, it argues that a simple productivity-price mechanism is not enough to explain the observed cross-country unit value patterns.

## 4.3 Across-product specialization test

The first empirical test tracks whether low- and high-wage countries export the same product codes to the United States.

- The main outcome is the share of products classified as *LMH*.
- If old trade theory operates mainly across products, the *LMH* share should be small.
- Figure III shows that the *LMH* share rises from about 30% in 1972 to about 62% in 1994.
- Table III shows that this pattern is robust to alternative cutoffs, excluding small shipments, and alternative definitions of low- and high-wage countries.

**Interpretation:** The paper rejects across-product specialization. Globalization increasingly puts high- and low-wage countries inside the same product categories.

#### 4.4 Within-product unit-value regressions

For each *LMH* product  $p$  and year  $t$ , the paper estimates

$$\log(u_{pct}) = \alpha_{pt} + \beta_{pt} \log(PCGDP_{ct}) + \varepsilon_{pct},$$

where  $u_{pct}$  is the unit value of product  $p$  from country  $c$  in year  $t$ .

- The paper estimates this regression separately for each product-year.
- Table IV reports the share of product-years in which  $\beta_{pt}$  is positive and significant.
- The share is generally large and increases over time.
- The relation is strongest for products imported from many countries, suggesting the result is not driven by thin markets.

**Interpretation:** Within the same product category, richer countries tend to export higher-unit-value varieties.

#### 4.5 Endowment regressions

The paper then estimates panel regressions of the form

$$\log(u_{pct}) = \alpha_{pt} + \beta Z_{ct} + \varepsilon_{pct},$$

where  $Z_{ct}$  is exporter per capita GDP, capital per labor, or skill per labor.

- Product-year fixed effects compare exporters within the same product-year.
- Exporter characteristics are positively related to unit values.
- Table V implies that 10% increases in exporter per capita GDP, capital per labor, and skill per labor are associated with roughly 1.3%, 4.4%, and 5.0% increases in unit values, respectively.

**Interpretation:** The factor-abundance result is not just a rich-country dummy. It is directly connected to capital and skill abundance.

#### 4.6 Industry-specific and production-technique regressions

For manufacturing industry  $i$ , the paper estimates

$$\log(u_{pict}) = \alpha_{pt} + \beta_i K/L_{ct} + \varepsilon_{pict}.$$

It also estimates, for 1990,

$$\log(u_{pic}) = \alpha_p + \beta_i k_{ic} + \varepsilon_{pic},$$

where  $k_{ic}$  is the capital intensity of industry  $i$  in country  $c$ .

- Tables VI and VII show that the capital-per-labor gradient is positive across major manufacturing industries.
- Table VIII shows that product unit values are positively related to exporter-industry capital intensity in most manufacturing industries.

- These results connect product prices to actual production techniques, not only to country income.

**Interpretation:** Higher-unit-value varieties appear to be produced with more capital-intensive techniques. This is exactly what a within-product factor-proportions interpretation predicts.

## 4.7 Relative factor accumulation over time

The paper estimates

$$\Delta u_{pc} = \alpha + \beta \Delta Z_c + \varepsilon_{pc},$$

where  $\Delta u_{pc}$  is the change in exporter-product unit value relative to the mean change across exporters of the same product, and  $\Delta Z_c$  is the change in exporter relative income, capital per worker, or skill per worker.

- Table IX uses the constant set of products exported in both 1972 and 1988.
- Countries whose relative income, capital abundance, or skill abundance rises tend to see relative unit values rise.
- The constant-sample restriction is severe, so the paper treats the evidence as suggestive but supportive.

**Interpretation:** Within-product specialization is dynamic. As countries accumulate capital and skill, they appear to move up the within-product quality or sophistication ladder.

## 5 Identification and empirical design

### 5.1 What the design can identify

- The paper identifies systematic associations between exporter characteristics and unit values within narrowly defined product categories.
- It does not identify a fully causal effect of factor abundance on product quality.
- The empirical design is strongest as a test of whether product-level trade patterns are consistent with the predictions of old and new trade theory.

**Interpretation:** The paper is not a randomized experiment. Its strength comes from extremely detailed product-country variation and multiple mutually reinforcing tests.

### 5.2 Across-product versus within-product logic

- Across-product tests ask whether low- and high-wage countries export the same product codes.
- Within-product tests ask whether, conditional on the same product code, high-wage exporters sell higher-unit-value varieties.
- The paper finds no support for across-product specialization, but strong support for within-product specialization.

**Interpretation:** This is the paper’s main conceptual distinction. Standard industry or product classifications can hide factor-proportions specialization.



### 5.3 Role of unit values

- Unit values are used as proxies for product attributes, quality, sophistication, or vertical differentiation.
- Because unit values may also contain measurement error, the paper checks whether the patterns are systematic across products, industries, exporter endowments, production techniques, and time.
- The repeated positive relationships make it unlikely that the result is only coding error.

**Interpretation:** The paper’s interpretation is most convincing when the unit value patterns line up simultaneously with country factor abundance and industry input intensity.

### 5.4 Alternative explanations and limits

- **Transfer pricing.** Multinationals may report transfer prices strategically. The paper notes that the direction of bias is ambiguous and difficult to control.
- **Bargaining power.** U.S. firms may bargain lower prices from low-wage exporters. The author argues that such persistent, large differences would be difficult to sustain as the full explanation.
- **Demand-side tastes.** U.S. consumers may prefer goods from high-income countries. If this reflects higher quality or added attributes, it is consistent with the factor-proportions interpretation.
- **Outsourcing.** Different stages of production could be classified inside the same product code. The paper checks that excluding products containing the word “part” does not change the results materially.

**Interpretation:** The paper is careful not to claim that unit values are a perfect quality measure. The central claim is that the full pattern of evidence is most naturally interpreted as vertical, factor-related within-product specialization.

## 6 Tables and figures

### 6.1 Figure I and Table I: theory and product mapping

**Read:**

- Figure I shows the multiple-cone Heckscher-Ohlin logic: countries with different capital-labor ratios occupy different cones and specialize in goods with matching input intensities.
- Table I maps coarse SITC industries to much finer HS products, showing how broad industry categories contain thousands of product codes.
- Manufactured materials, machinery, and miscellaneous manufactures contain especially many detailed product categories.

**Economic message:** Industry-level data are too coarse to detect the relevant margin of specialization. The paper’s product-level data allow the author to look inside industries and ask whether specialization occurs within product categories.

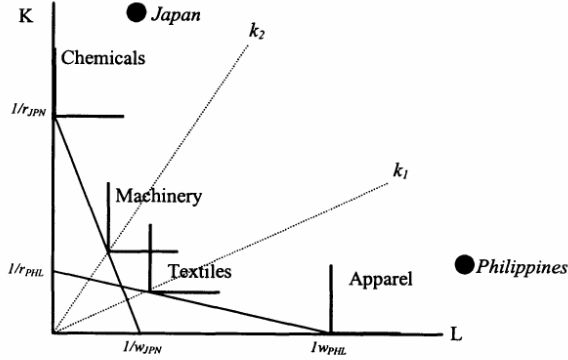


FIGURE I  
Multiple Cone Heckscher-Ohlin Specialization

(a) Figure I: Multiple-cone Heckscher-Ohlin specialization

TABLE I  
MAPPING PRODUCTS TO INDUSTRIES

| SITC1 industry               | Example of SITC2 industries                                | Example of HS product                             | Number of products (1994) |
|------------------------------|------------------------------------------------------------|---------------------------------------------------|---------------------------|
| 0 Food                       | Meat, Dairy Products, Fruit                                | Sheep, live                                       | 1645                      |
| 1 Beverages and Tobacco      | Wine, Cigarettes                                           | Carbonated soft drinks                            | 147                       |
| 2 Crude Materials            | Rubber, Cork, Wood, Textile Fibers                         | Silkworm cocoons suitable for reeling             | 787                       |
| 3 Mineral Fuels              | Coal, Coke, Petroleum, Natural Gas, Electric Current       | Unleaded gasoline                                 | 95                        |
| 4 Animal and Vegetable Oils  | Lard, Soybean Oil, Linseed Oil                             | Tallow, edible                                    | 77                        |
| 5 Chemicals                  | Organic Chemicals, Dyes, Medicines, Fertilizers, Plastics  | Chloroform                                        | 1886                      |
| 6 Manufactured Materials     | Leather, Textile Yarn, Paper, Steel, Cork Products         | Diaries and address books, of paper or paperboard | 4117                      |
| 7 Machinery                  | Power Generating Machinery, Computers, Autos               | Ultrasonic scanning apparatus                     | 2871                      |
| 8 Miscellaneous Manufactures | Apparel, Footwear, Plumbing, Scientific Equipment, Cameras | Boys' shorts cotton playsuit parts, not knit      | 3665                      |
| 9 Not Elsewhere Classified   | Special Transactions, Coins, Gold                          | Sound recordings for State Dept use               | 61                        |

(b) Table I: Mapping products to industries

## 6.2 Figure II and Table II, part 1: product resolution and country cohorts

Read:

- Figure II shows that broad industry cells are frequently nonzero, but product-level country-product cells are much sparser.
- This means industry-level data mechanically understate specialization because almost every country exports something in broad industries.
- Table II lists the countries used in the sample and shows when they are classified as low-wage under the baseline and alternative rules.

**Economic message:** Product-level detail reveals more meaningful trade participation. At the same time, the country cohort definitions make clear that the paper is comparing very different exporters, such as China, India, Japan, and high-income European countries.

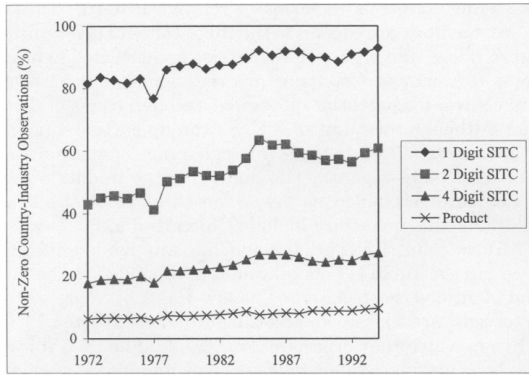


FIGURE II

Percent of Nonzero Country-Industry and Country-Product Observations in a Panel of U. S. Manufacturing Imports

The graph displays the percent of nonzero country-industry cells across all U. S. manufacturing imports for the noted level of aggregation. SITC is the Standard International Trade Classification; four-digit industries are more finely defined than one-digit industries. Product refers to the seven-digit TS or ten-digit HS product categories, which are the most detailed description of U. S. trade flows available. Product classification switched from TS to HS in 1989.

(a) Figure II: Nonzero observations by aggregation level

COUNTRIES IN DATASET

| Country       | 0–30th Percentile | <5% US PCGDP | Lone (%) |
|---------------|-------------------|--------------|----------|
| afghanistan*  | 72–81             | .            | 0.23     |
| albania       | 92–94             | 80–94        | 0.02     |
| algeria       | .                 | .            | .        |
| angola*       | 92–94             | 85–94        | 0.01     |
| argentina     | .                 | .            | .        |
| australia     | .                 | .            | .        |
| AUSTRIA       | .                 | .            | .        |
| bahamas       | .                 | .            | .        |
| bahrain*      | .                 | .            | .        |
| BANGLADESH    | 72–94             | 72–94        | 0.37     |
| barbados      | .                 | .            | .        |
| BELGIUM       | .                 | .            | .        |
| BELIZE        | .                 | .            | .        |
| benin         | 72–94             | 72–94        | 0.02     |
| bermuda       | .                 | .            | .        |
| bolivia*      | .                 | 79–94        | .        |
| brazil        | .                 | .            | .        |
| bulgaria*     | .                 | .            | .        |
| burkina faso  | 72–94             | 72–94        | 0.03     |
| burundi       | 72–94             | 72–94        | 0.01     |
| CAMBODIA*     | 89–94             | 87–94        | 0.01     |
| cameroon      | 72–80             | 72–94        | 0.05     |
| canada        | .                 | .            | .        |
| cen afr rep   | 72–94             | 72–94        | 0.02     |
| chad          | 72–94             | 72–94        | 0.01     |
| CHILE         | .                 | .            | .        |
| CHINA         | 72–94             | 72–94        | 54.74    |
| COLOMBIA      | .                 | .            | .        |
| congo dr      | 72–94             | 72–94        | 0.23     |
| congo r       | 77–78             | 72–81;86–94  | .        |
| costa rica    | .                 | .            | .        |
| cote d’ivoire | 85                | 80–94        | 0.03     |
| cyprus*       | .                 | .            | .        |
| czech rep*    | .                 | .            | .        |
| denmark       | .                 | .            | .        |
| dominican rep | .                 | .            | .        |
| ECUADOR       | .                 | .            | .        |
| EGYPT         | 72–86;89          | 72–94        | 0.68     |
| el salvador   | 81–83             | .            | 0.13     |
| eq guinea*    | 87–94             | 85–94        | .        |
| ethiopia*     | 83–94             | 81–94        | 0.07     |
| fiji          | .                 | .            | .        |
| finland       | .                 | .            | .        |
| france        | .                 | .            | .        |
| gabon         | .                 | .            | .        |
| gambia        | 72–94             | 72–94        | 0.06     |
| GERMANY       | .                 | .            | .        |
| ghana         | 72–94             | 72–94        | 0.36     |

(b) Table II, part 1: Countries in dataset

### 6.3 Table II, parts 2 and 3: country cohort details

Read:

- The continuation pages of Table II show the full set of countries and their classification histories.
- Large low-wage exporters such as China, India, Pakistan, and the Philippines are important for the *LMH* classification.
- The table also reports which single low-wage countries define some *LMH* products, a concern addressed in robustness checks.

**Economic message:** The *LMH* finding is not based on a vague high-low comparison. It is built from a country-by-year classification that the paper then stress-tests with alternative cutoffs.

TABLE II  
CONTINUED

| Country        | 0–30th percentile       | <5% US PCGDP            | Lone (%) |
|----------------|-------------------------|-------------------------|----------|
| greece         | .                       | .                       | .        |
| greenland*     | .                       | .                       | .        |
| guatemala      | .                       | .                       | .        |
| guinea*        | 88–94                   | 86–94                   | 0.03     |
| guinea-bissau  | 72–94                   | 72–94                   | 0.06     |
| guyana         | 82–94                   | 72–94                   | 0.27     |
| haiti          | 72–94                   | 72–94                   | 2.44     |
| honduras       | 73;75–77;81–83;90–92;94 | 72–94                   | 0.63     |
| HONG KONG      | .                       | .                       | .        |
| hungary*       | .                       | .                       | .        |
| ICELAND        | .                       | .                       | .        |
| INDIA          | 72–94                   | 72–94                   | 21.65    |
| INDONESIA      | 72–91                   | 72–94                   | 2.49     |
| iran*          | .                       | 88–90                   | .        |
| IRELAND        | .                       | .                       | .        |
| israel         | .                       | .                       | .        |
| ITALY          | .                       | .                       | .        |
| jamaica        | .                       | .                       | .        |
| JAPAN          | .                       | .                       | .        |
| jordan*        | .                       | .                       | .        |
| kenya          | 72–94                   | 72–94                   | 0.66     |
| kiribati*      | 87–89                   | 88–90                   | 0.04     |
| KOREA          | .                       | .                       | .        |
| kuwait         | .                       | .                       | .        |
| lao pdr*       | 90–94                   | 84–94                   | 0.01     |
| lebanon*       | .                       | .                       | .        |
| liberia*       | 73;78–87                | .                       | 0.09     |
| madagascar     | 72–94                   | 72–94                   | 0.10     |
| malawi         | 72–94                   | 72–94                   | 0.06     |
| MALAYSIA       | .                       | .                       | .        |
| mali           | 72–94                   | 72–94                   | 0.22     |
| MALTA          | .                       | .                       | .        |
| mauritania     | 72–94                   | 72–94                   | 0.05     |
| mauritius      | .                       | .                       | .        |
| mexico         | .                       | .                       | .        |
| MOROCCO        | 85–87                   | 72–74;81;84;87;89;92–93 | 0.31     |
| mozambique*    | 82–94                   | 80–94                   | 0.14     |
| nepal          | 72–94                   | 72–94                   | 0.25     |
| neth antilles* | .                       | .                       | .        |
| netherlands    | .                       | .                       | .        |
| new caledonia* | .                       | .                       | .        |
| new zealand    | .                       | .                       | .        |
| nicaragua      | 79–82;89–94             | 72–73;78–94             | 0.28     |
| niger          | 72–94                   | 72–94                   | 0.37     |
| nigeria        | 72–75;83–94             | 72–94                   | 0.30     |
| NORWAY         | .                       | .                       | .        |

TABLE II  
CONTINUED

| Country            | 0–30th percentile | <5% US PCGDP | Lone (%) |
|--------------------|-------------------|--------------|----------|
| OMAN               | .                 | .            | .        |
| PAKISTAN           | 72–94             | 72–94        | 2.62     |
| panama             | .                 | .            | .        |
| papua new guinea   | .                 | 76–94        | .        |
| PARAGUAY           | .                 | .            | .        |
| peru               | .                 | .            | .        |
| philippines        | 72–89;91          | 84–94        | 7.77     |
| poland*            | .                 | .            | .        |
| PORTUGAL           | .                 | .            | .        |
| qatar              | .                 | .            | .        |
| romania*           | .                 | .            | .        |
| rwanda             | 72–94             | 72–94        | 0.02     |
| saudi arabia       | .                 | .            | .        |
| senegal            | 72–91;94          | 72–94        | 0.17     |
| SEYCHELLES         | .                 | .            | .        |
| sierra leone       | 72–94             | 72–94        | 0.10     |
| SINGAPORE          | .                 | .            | .        |
| somalia*           | 72–90             | .            | 0.03     |
| south africa       | .                 | .            | .        |
| SPAIN              | .                 | .            | .        |
| SRI LANKA          | 72–93             | 72–94        | 0.59     |
| st. kitts & nevis* | .                 | .            | .        |
| sudan              | 72–94             | 72–94        | 0.05     |
| suriname           | .                 | 72–75;79–94  | .        |
| sweden             | .                 | .            | .        |
| switzerland        | .                 | .            | .        |
| SYRIA              | .                 | 72–94        | .        |
| TAIWAN             | .                 | 72–77        | .        |
| tanzania*          | 90–94             | 88–94        | 0.04     |
| THAILAND           | 72–78             | 73–77        | 0.96     |
| togo               | 72–94             | 72–94        | 0.03     |
| trinidad & tobago  | .                 | .            | .        |
| TUNISIA            | .                 | .            | .        |
| turkey             | .                 | .            | .        |
| uae*               | .                 | .            | .        |
| uganda*            | 84–94             | 82–94        | 0.01     |
| united kingdom     | .                 | .            | .        |
| uruguay            | .                 | .            | .        |
| venezuela          | .                 | .            | .        |
| vietnam*           | 93–94             | 84–94        | 0.02     |
| zambia             | 78–94             | 72–94        | 0.05     |
| zimbabwe           | 86                | 72–94        | 0.02     |

First column indicates years for which the country's PCGDP is within the thirtieth percentile of the world distribution. Second column of each panel indicates years for which the country's PCGDP is less than 5 percent of U. S. PCGDP. Third column of each panel indicates percent of single low-wage country LMH products, if any, defined solely by the noted country (see text). An asterisk after country name indicates PCGDP for some years is missing. Uppercase (lowercase) letters indicate that the country's PCGDP relative to the U. S. increases (decreases) between 1972 and 1994.

(a) Table II, part 2: Countries in dataset, continued

(b) Table II, part 3: Countries in dataset, continued

## 6.4 Figure III and Table III: across-product specialization

Read:

- Figure III classifies U.S. import products by source-country income groups.
- The *LMH* category rises from about 30% of import products in 1972 to about 62% in 1994.
- Table III shows that this increase survives several robustness checks: removing small shipments, changing percentile cutoffs, excluding products defined by one low-wage country, and using income relative to the United States.

**Economic message:** The United States increasingly imports the same products from both low-wage and high-wage countries. This rejects the simple view that factor-proportions specialization mainly occurs across product categories.

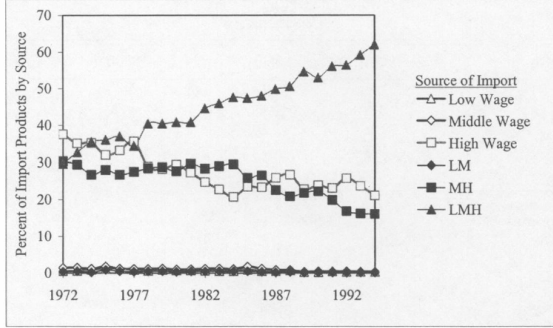


FIGURE III

**Breakdown of U. S. Imports by Exporter per Capita GDP Level, 1972–1994**  
U. S. import products are classified according to the income level of the exporting country. Exporters are classified as low-, middle-, or high-wage if their per capita GDP is below the thirtieth, between the thirtieth and seventieth or above the seventieth percentiles of world PCGDP, respectively. Low-wage products (L) originate solely in low-wage countries (e.g., China), while LMH products originate simultaneously in at least one low- and one high-wage country. The six product classifications in the figure are mutually exclusive.

(a) Figure III: Breakdown of imports by exporter income level

TABLE III  
SENSITIVITY ANALYSIS OF ACROSS-PRODUCT SPECIALIZATION RESULTS

| Row | PCGDP percentile cutoffs<br>(Figure III) | Additional restrictions                                                                                                                                   | Percent of LMH products in U. S. imports |      |
|-----|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|------|
|     |                                          |                                                                                                                                                           | 1972                                     | 1994 |
| 1   | 30 <sup>th</sup> –70 <sup>th</sup>       | None                                                                                                                                                      | 30                                       | 62   |
| 2   | 30 <sup>th</sup> –70 <sup>th</sup>       | Exclude exporter-product observations where export value is less than \$10,000                                                                            | 15                                       | 54   |
| 3   | 40 <sup>th</sup> –60 <sup>th</sup>       | None                                                                                                                                                      | 45                                       | 65   |
| 4   | 20 <sup>th</sup> –80 <sup>th</sup>       | None                                                                                                                                                      | 25                                       | 34   |
| 5   | 30 <sup>th</sup> –90 <sup>th</sup>       | None                                                                                                                                                      | 27                                       | 61   |
| 6   | 30 <sup>th</sup> –70 <sup>th</sup>       | Exclude LMH products exported by just one low-wage country. (See Table II for list of countries)                                                          | 17                                       | 44   |
| 7   | 30 <sup>th</sup> –70 <sup>th</sup>       | Exclude LHM products where the only low-wage exporter is China.                                                                                           | 28                                       | 49   |
| 8   | PCGDP relative to the U. S.              | Countries are low wage if their PCGDP relative to the U. S. is less than 5%, and high wage if it is greater than 95%.                                     | 40                                       | 62   |
| 9   | PCGDP relative to the U. S.              | Countries are low wage if their PCGDP relative to the U. S. is less than 2%, and high wage if it is greater than 98%                                      | 38                                       | 57   |
| 10  | PCGDP relative to the U. S.              | Use 5%–95% cutoff. Exclude low-wage countries whose PCGDP rises relative to the U. S. Exclude high-wage countries whose PCGDP falls relative to the U. S. | 9                                        | 22   |

This table reports the share of U. S. import products sourced simultaneously from low-, middle-, and high-wage countries in 1972 and 1994 according to different robustness tests. Per capita GDP percentile splits refer to the breakpoints used to classify countries as low, middle, and high wage. Products are classified according to the set of countries in which they originate.

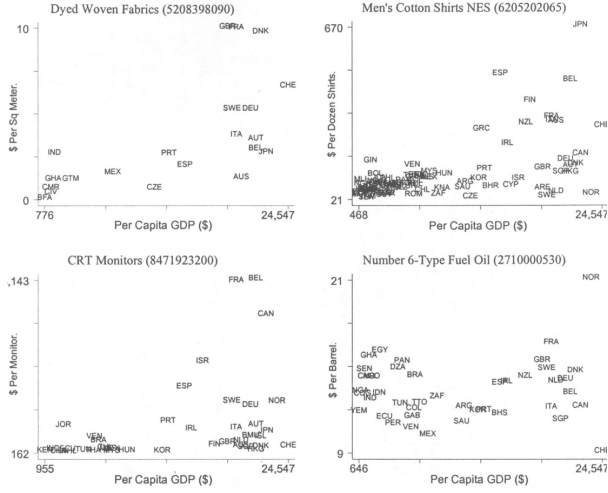
(b) Table III: Sensitivity analysis of across-product specialization

## 6.5 Figure IV and Table IV: within-product unit values and exporter income

Read:

- Figure IV plots unit values against exporter per capita GDP for four products: dyed woven fabrics, men's cotton shirts, CRT monitors, and fuel oil.
- The manufactured products show positive unit value-income relationships; the natural-resource product does not show the same pattern.
- Table IV formalizes the pattern across all *LMH* manufacturing products and years. More broadly sourced products are especially likely to show a positive unit value-PCGDP relationship.

**Economic message:** Within the same product code, richer exporters tend to sell higher-unit-value varieties. The pattern is strongest for manufactured products where vertical differentiation is plausible.



(a) Figure IV: Unit values versus exporter per capita GDP

TABLE IV  
UNIT VALUE AND EXPORTER PER CAPITA GDP, BY PRODUCT AND YEAR

| Year | Percent of LMH manufacturing products with significant positive relationship between unit value and exporter PCGDP |                                          |                                          |
|------|--------------------------------------------------------------------------------------------------------------------|------------------------------------------|------------------------------------------|
|      | All LMH products                                                                                                   | LMH products with more than 20 exporters | LMH products with more than 40 exporters |
| 1972 | 43                                                                                                                 | 55                                       | 61                                       |
| 1973 | 45                                                                                                                 | 57                                       | 69                                       |
| 1974 | 40                                                                                                                 | 53                                       | 69                                       |
| 1975 | 41                                                                                                                 | 57                                       | 68                                       |
| 1976 | 43                                                                                                                 | 54                                       | 55                                       |
| 1977 | 42                                                                                                                 | 55                                       | 53                                       |
| 1978 | 43                                                                                                                 | 53                                       | 55                                       |
| 1979 | 45                                                                                                                 | 56                                       | 60                                       |
| 1980 | 46                                                                                                                 | 56                                       | 50                                       |
| 1981 | 42                                                                                                                 | 53                                       | 59                                       |
| 1982 | 43                                                                                                                 | 50                                       | 61                                       |
| 1983 | 43                                                                                                                 | 49                                       | 55                                       |
| 1984 | 43                                                                                                                 | 50                                       | 57                                       |
| 1985 | 45                                                                                                                 | 56                                       | 61                                       |
| 1986 | 47                                                                                                                 | 56                                       | 68                                       |
| 1987 | 49                                                                                                                 | 58                                       | 72                                       |
| 1988 | 52                                                                                                                 | 62                                       | 71                                       |
| 1989 | 49                                                                                                                 | 60                                       | 74                                       |
| 1990 | 52                                                                                                                 | 63                                       | 75                                       |
| 1991 | 51                                                                                                                 | 63                                       | 74                                       |
| 1992 | 52                                                                                                                 | 63                                       | 76                                       |
| 1993 | 50                                                                                                                 | 61                                       | 74                                       |
| 1994 | 49                                                                                                                 | 60                                       | 73                                       |

This table reports the number of manufacturing products exhibiting a positive and significant correlation (at the 10 percent level) between product unit value and exporter PCGDP. The first column reports significance across the full sample of LMH products. Subsequent columns restrict the sample to LMH products imported from the indicated number of countries. Sample size declines across columns: in 1994 there are 5528, 2229, and 447 LMH products in the full,  $n > 20$  and  $n > 40$  samples, respectively.

(b) Table IV: Unit value and exporter PCGDP by product-year

## 6.6 Tables V and VI: exporter characteristics and industry-level results

Read:

- Table V regresses log unit values on exporter per capita GDP, capital per labor, and skill per labor with product-year fixed effects.
- All three exporter characteristics are positively related to unit values.
- Table VI repeats the capital-per-labor test by broad manufacturing industry. The coefficient is positive and statistically significant in chemicals, manufactured materials, machinery, and miscellaneous manufactures.

**Economic message:** The within-product price gradient is directly related to country factor abundance. The evidence is not confined to one industry group.

TABLE V  
UNIT VALUES AND EXPORTER CHARACTERISTICS, 1972–1994

| Regressor                         | Log unit value | Log unit value | Log unit value |
|-----------------------------------|----------------|----------------|----------------|
| Log PCGDP                         | 0.134***       |                |                |
|                                   | 0.037          |                |                |
| Log capital per labor (\$000)     |                | 0.435***       |                |
|                                   |                | 0.065          |                |
| Log skill per labor               |                |                | 0.501**        |
|                                   |                |                | 0.089          |
| Product-year dummies              | Yes            | Yes            | Yes            |
| Product-country-year observations | 214,852        | 214,852        | 214,852        |
| Number of unique products         | 12,024         | 12,024         | 12,024         |
| Number of unique countries        | 48             | 48             | 48             |
| $R^2$                             | 0.77           | 0.78           | 0.77           |

This table reports OLS regression results of exporter unit value on real exporter PCGDP, real exporter capital per worker, and exporter skill abundance across LMH products (see text). Sample restricted to available data across independent variables. GDP data are from the World Bank (2000). Capital per labor ratios are from Penn World Tables 5.6; 1992 values are used for 1994. Education attainment data are from Barro and Lee (2000); 1970 and 1995 data are used for 1972 and 1994, respectively. Robust standard errors adjusted for exporter clustering are noted below coefficients. Results for fixed effects are suppressed. \*\*\*, \*\*, and \* refer to statistical significance at the 1 percent, 5 percent, and 10 percent levels, respectively.

(a) Table V: Unit values and exporter characteristics

TABLE VI  
UNIT VALUES AND EXPORTER CHARACTERISTICS, BY SITC1 INDUSTRY

|                                   | Chemicals<br>SITC 5 | Manufactured<br>materials<br>SITC 6 | Machinery<br>SITC 7 | Miscellaneous<br>manufactures<br>SITC 8 |
|-----------------------------------|---------------------|-------------------------------------|---------------------|-----------------------------------------|
| Regressor                         | Log(unit value)     | Log(unit value)                     | Log(unit value)     | Log(unit value)                         |
| Log(K/L)                          | 0.209***            | 0.282***                            | 0.539***            | 0.413***                                |
|                                   | 0.076               | 0.041                               | 0.118               | 0.051                                   |
| Product-year dummies              | Yes                 | Yes                                 | Yes                 | Yes                                     |
| Product-country-year observations | 61,771              | 313,115                             | 144,982             | 429,478                                 |
| Number of unique products         | 1,543               | 5,706                               | 2,932               | 7,523                                   |
| Number of unique countries        | 59                  | 59                                  | 59                  | 59                                      |
| $R^2$                             | 0.59                | 0.71                                | 0.78                | 0.78                                    |

This table displays OLS coefficients from a panel regression of exporter-product unit values on log exporter capital per labor across LMH products by SITC1 industry from 1972 to 1994. Robust standard errors adjusted for exporter clustering are listed below each coefficient. Results for fixed effects are suppressed. \*\*\*, \*\*, and \* refer to statistical significance at the 1 percent, 5 percent, and 10 percent levels, respectively.

(b) Table VI: Unit values and exporter characteristics by SITC1 industry

## 6.7 Tables VII and VIII: constant sample and production techniques

Read:

- Table VII restricts the sample to products and trading partners that exist in both 1972 and 1988. The positive capital-per-labor gradient remains.
- Table VIII relates product unit values in 1990 to exporter-industry capital intensity.
- The coefficient is positive and statistically significant in 26 of 28 manufacturing industries, with large slopes in industries such as apparel, machinery, and electrical machinery.

**Economic message:** The results survive a severe constant-sample check and line up with actual production techniques. This is strong support for within-product factor-proportions specialization.

TABLE VII  
UNIT VALUES AND EXPORTER CHARACTERISTICS (CONSTANT SAMPLE),  
BY SITC1 INDUSTRY

| Regressor                            | Chemicals<br>SITC 5 | Manufactured<br>Materials<br>SITC 6 | Machinery<br>SITC 7 | Miscellaneous<br>Manufactures<br>SITC 8 |
|--------------------------------------|---------------------|-------------------------------------|---------------------|-----------------------------------------|
|                                      | Log(unit<br>value)  | Log(unit<br>value)                  | Log(unit<br>value)  | Log(unit<br>value)                      |
| Log( $K/L$ )                         | 0.244***<br>0.046   | 0.247***<br>0.034                   | 0.578***<br>0.139   | 0.328***<br>0.054                       |
| Product-year<br>dummies              | Yes                 | Yes                                 | Yes                 | Yes                                     |
| Product-country-year<br>observations | 6,412               | 66,151                              | 5,205               | 72,946                                  |
| Number of unique<br>products         | 54                  | 459                                 | 67                  | 522                                     |
| Number of unique<br>countries        | 56                  | 59                                  | 55                  | 59                                      |
| $R^2$                                | 0.59                | 0.78                                | 0.83                | 0.80                                    |

This table displays OLS coefficients from a panel regression of exporter-product unit values on log exporter capital per labor across LMH products by SITC1 industry from 1972 to 1988. Sample is restricted to trading partners and products existing in 1972 and 1988. Robust standard errors adjusted for exporter clustering are listed below each coefficient. Results for fixed effects are suppressed. \*\*\*, \*\*, and \* refer to statistical significance at the 1 percent, 5 percent, and 10 percent levels, respectively.

(a) Table VII: Constant-sample exporter characteristics

TABLE VIII  
PRODUCT UNIT VALUES AND INDUSTRY PRODUCTION TECHNIQUES, 1990

| ISIC Industry (Rev 2)                     | Slope   | $R^2$ | Product-country<br>observations |
|-------------------------------------------|---------|-------|---------------------------------|
| 311 Food products                         | 0.35*** | 0.49  | 8,505                           |
| 313 Beverages                             | 0.24*** | 0.46  | 546                             |
| 314 Tobacco                               | 0.05    | 0.30  | 176                             |
| 321 Textiles                              | 0.62*** | 0.43  | 14,440                          |
| 322 Wearing apparel, except footwear      | 0.81*** | 0.49  | 8,276                           |
| 323 Leather products                      | 0.37*** | 0.44  | 863                             |
| 324 Footwear, except rubber or plastic    | 0.58*** | 0.45  | 2,177                           |
| 331 Wood products, except furniture       | 0.29*** | 0.55  | 1,053                           |
| 332 Furniture, except metal               | 0.33*** | 0.33  | 547                             |
| 341 Paper and products                    | 0.03    | 0.47  | 1,036                           |
| 342 Printing and publishing               | 0.18*** | 0.64  | 1,393                           |
| 351 Industrial chemicals                  | 0.30*** | 0.49  | 4,192                           |
| 352 Other chemicals                       | 0.45*** | 0.53  | 2,123                           |
| 353 Petroleum refineries                  | 0.18**  | 0.58  | 310                             |
| 354 Miscellaneous petroleum and coal      | 0.70*** | 0.66  | 196                             |
| 355 Rubber products                       | 0.39*** | 0.48  | 1,108                           |
| 356 Plastic products                      | 0.43*** | 0.56  | 2,173                           |
| 361 Pottery, china, earthenware           | 0.35*** | 0.56  | 1,070                           |
| 362 Glass and products                    | 0.50*** | 0.51  | 1,296                           |
| 369 Other nonmetallic mineral products    | 0.27*** | 0.48  | 1,406                           |
| 371 Iron and steel                        | 0.66*** | 0.64  | 2,140                           |
| 372 Nonferrous metals                     | 0.67*** | 0.73  | 852                             |
| 381 Fabricated metal products             | 0.47*** | 0.54  | 5,736                           |
| 382 Machinery, except electrical          | 0.46*** | 0.49  | 7,541                           |
| 383 Machinery, electric                   | 0.85*** | 0.45  | 3,134                           |
| 384 Transport equipment                   | 0.33*** | 0.50  | 3,864                           |
| 385 Professional and scientific equipment | 0.52*** | 0.37  | 2,813                           |
| 390 Other manufactured products           | 0.30*** | 0.56  | 3,392                           |

This table reports OLS regression results of cross-country regressions of log country-product unit value on log country-ISIC industry capital intensity, by ISIC industry. All data are for 1990. Regressions include product fixed effects. Industry-country capital intensities are from Schott [2003]. Products are concorded to industries using Maskus' [1991] SITC to ISIC industry concordance and the product to SITC concordance provided by Feenstra [1996]. Results for constant are suppressed. \*\*\*, \*\*, and \* refer to statistical significance at the 1 percent, 5 percent, and 10 percent levels, respectively.

(b) Table VIII: Unit values and production techniques

## 6.8 Table IX: relative factor accumulation over time

Read:

- Table IX studies changes between 1972 and 1988 for a constant set of *LMH* products.
- Relative increases in exporter per capita GDP, capital per worker, and skill per worker are positively associated with relative increases in exporter-product unit values.
- The evidence is based on a restricted sample, so the paper treats it as supportive rather than definitive.

**Economic message:** Countries appear to move up the within-product quality ladder as they accumulate capital and skill. This dynamic result reinforces the cross-sectional evidence.



TABLE IX  
UNIT VALUE CHANGES AND RELATIVE FACTOR ACCUMULATION, 1972 TO 1988

| Regressor                                           | Change in<br>exporter-<br>product unit<br>value | Change in<br>exporter-<br>product unit<br>value | Change in<br>exporter-<br>product unit<br>value |
|-----------------------------------------------------|-------------------------------------------------|-------------------------------------------------|-------------------------------------------------|
| Change in exporter's relative<br>PCGDP              | 0.476**<br>0.232                                |                                                 |                                                 |
| Change in exporter's relative<br>capital per worker |                                                 | 0.463*<br>0.246                                 |                                                 |
| Change in exporter's relative<br>skill per worker   |                                                 |                                                 | 0.006*<br>0.003                                 |
| Product-country observations                        | 2111                                            | 1456                                            | 1897                                            |
| Number of unique countries                          | 91                                              | 49                                              | 82                                              |
| $R^2$                                               | 0.30                                            | 0.32                                            | 0.30                                            |

This table reports OLS estimation results across the constant set of LMH products exported to the United States in both 1972 and 1988. Dependent variable is log difference of exporter-product unit value between 1972 and 1988. First regressor is change in exporter PCGDP percentile between 1972 and 1988. Second regressor is change in exporter real capital per worker (from Penn World Tables Mark 5.6) between 1972 and 1992. Third regressor is change in exporter secondary and higher education attainment [Barro and Lee 2000] between 1970 and 1990. All changes are relative to the mean change for the product. All countries do not export all products. Robust standard errors adjusted for country clustering are noted below coefficients. Results for the constant are suppressed. \*\*\*, \*\*, and \* refers to statistical significance at the 1 percent, 5 percent, and 10 percent levels, respectively.

Table IX: Unit value changes and relative factor accumulation

## 7 Short summary

- The paper studies product-level U.S. manufacturing imports from 1972 to 1994.
- It asks whether international specialization occurs across products or within products.
- Across-product specialization declines: low- and high-wage countries increasingly export the same product codes to the United States.
- Within-product specialization is strong: high-income, capital-abundant, and skill-abundant countries export higher-unit-value varieties within the same product categories.
- Unit values are also positively related to exporter-industry capital intensity, connecting prices to production techniques.
- Relative unit values rise when exporters accumulate income, capital, and skill relative to other exporters of the same products.
- The evidence supports a factor-proportions interpretation inside product categories.
- The evidence is difficult to reconcile with simple new trade models in which higher productivity lowers prices.
- The main implication is that globalization should be analyzed at the variety level, not just at the industry or product level.

## 8 Conclusion

### 8.1 Core conclusion

Schott shows that U.S. trading partners do not specialize across product categories in the simple Heckscher-Ohlin sense. Low- and high-wage countries increasingly export the same products to the United States. But this does not mean factor-proportions forces are absent. Instead, they appear inside product categories: high-wage, capital-abundant, and skill-abundant countries export higher-unit-value varieties of the same products.

### 8.2 Economic intuition

The intuition is that a product code such as “men’s cotton shirts” can include both low-quality labor-intensive shirts and high-quality, feature-rich, fashion-intensive shirts. Low-wage countries may specialize in the former, while high-wage countries specialize in the latter. Product codes therefore hide an important vertical margin of trade.

### 8.3 Incremental contribution revisited

The paper’s key contribution is to shift attention from industries to varieties. It shows that previous industry-level tests of factor-proportions trade may miss the relevant margin of specialization. It also provides a unified empirical framework for comparing old and new trade theory using product-level unit values.

### 8.4 Implications for globalization

The results imply that workers and firms in high-wage countries may not be in direct competition with low-wage producers even inside the same broad industry. If varieties are vertically differentiated, high-wage countries can remain active in apparel, textiles, electronics, and other industries by moving toward higher-quality or more sophisticated varieties. The degree of insulation depends on how substitutable these varieties are.

### 8.5 Implications for productivity and measurement

The paper also matters for productivity measurement. If product-level price differences reflect quality and input intensity, then aggregate or industry deflators may miss important variation across countries. This can bias comparisons of total factor productivity and technology gaps between developed and developing countries.

### 8.6 Limitations

The analysis relies on unit values, which are imperfect proxies for quality. Customs data can contain misclassification and quantity reporting errors. Transfer pricing, bargaining power, taste differences, and outsourcing may also affect observed prices. The evidence is therefore not a clean causal estimate of factor abundance on product quality. Its strength is the consistency of patterns across income, endowments, production techniques, and time.

## 8.7 Takeaway

The enduring takeaway is that international trade specialization is more subtle than the industry-level view suggests. Countries may converge in the product codes they export while diverging in the varieties they produce. Schott's paper shows that the old factor-proportions logic remains empirically powerful once we look inside product categories.